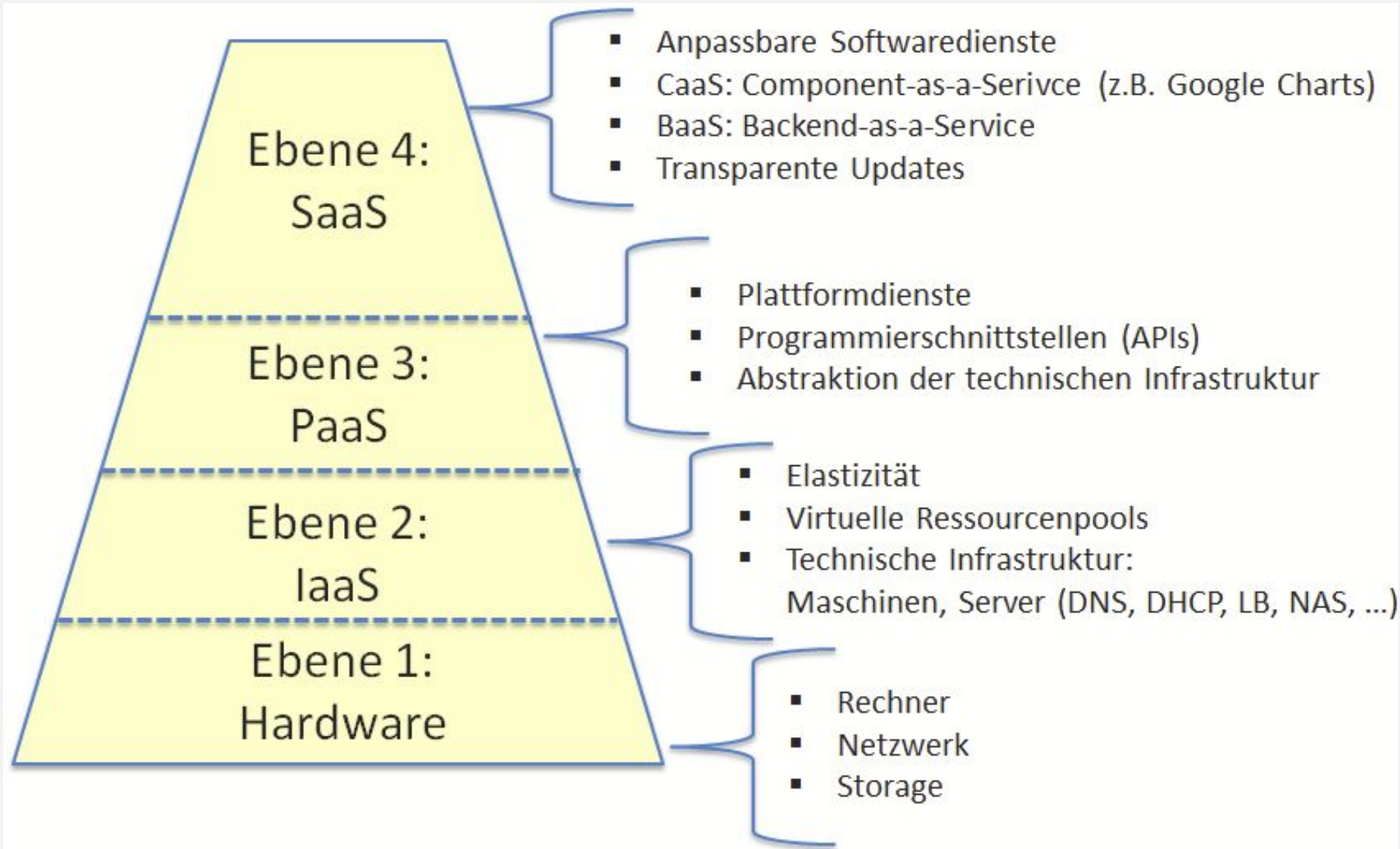




Cloud Computing Platform as a Service

Bernhard Schaidhammer
26.06.2026

PaaS im Schichtenmodell



Zielgruppe: User

Zielgruppe: Entwickler

Zielgruppe: Betrieb

Warum PaaS? Warum reicht uns nicht IaaS?

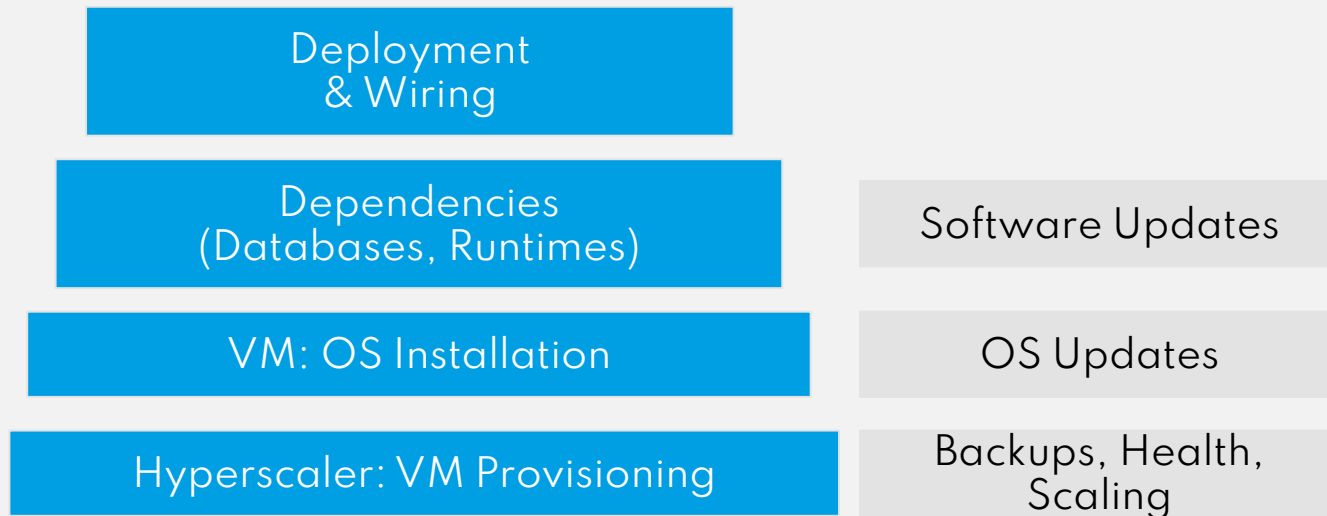
Stovepipe Architektur als Anti-Pattern



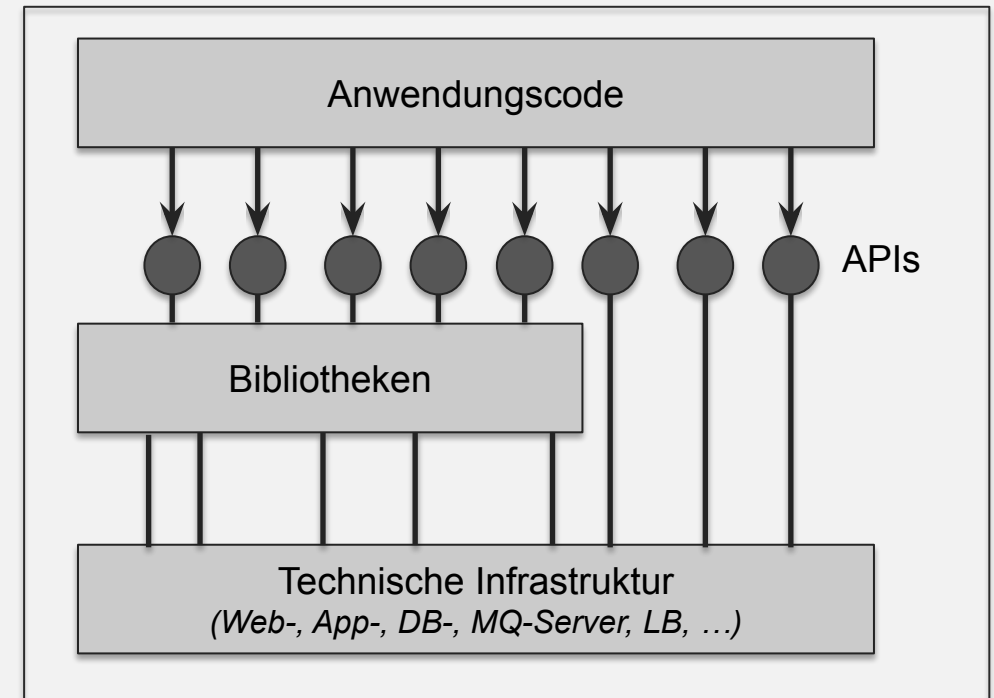
QA|WARE



Warum PaaS? Warum reicht uns nicht IaaS?

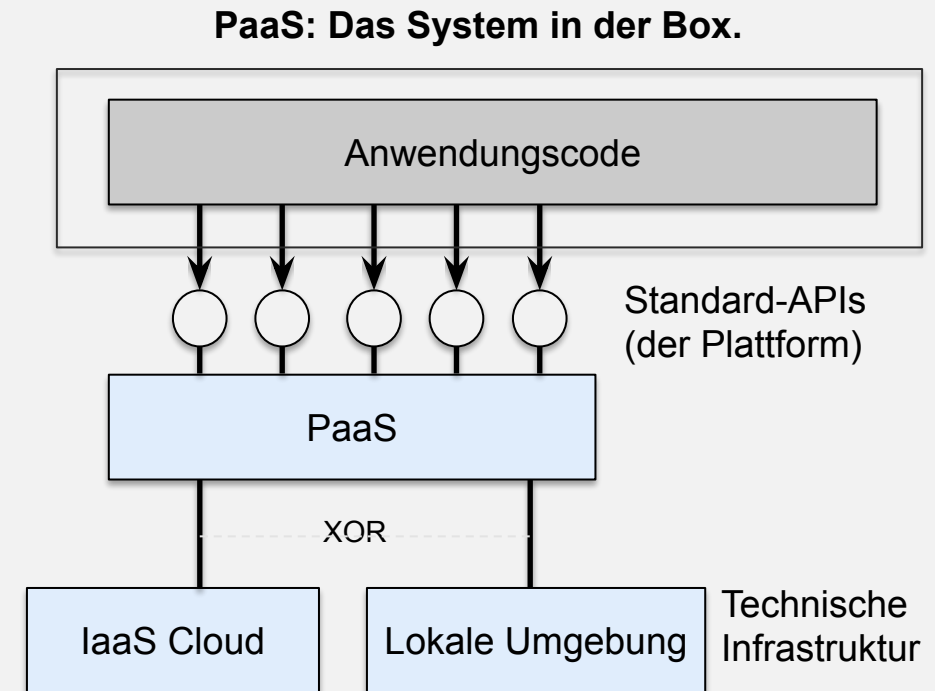


IaaS: Das System: Mühevoll verdrahtet.



Das Schichtenmodell des Cloud Computing: Vom Blech zur Anwendung.

- Die Anwendung wird per Applikationspaket oder als Quellcode deployed. Es ist kein Image mit Technischer Infrastruktur notwendig.
- Die Anwendung sieht nur Programmier- oder Zugriffsschnittstellen seiner Laufzeitumgebung.
„Engine and Operating System should not matter...“
- Es erfolgt eine automatische Skalierung der Anwendung.
- Entwicklungswerkzeuge (insb. Plugins für IDEs und Buildsysteme sowie eine lokale Testumgebung) stehen zur Verfügung.
„Deploy to Cloud“
- Die Plattform bietet eine Schnittstelle zur Administration und zum Monitoring der Anwendungen.



PaaS: Definition

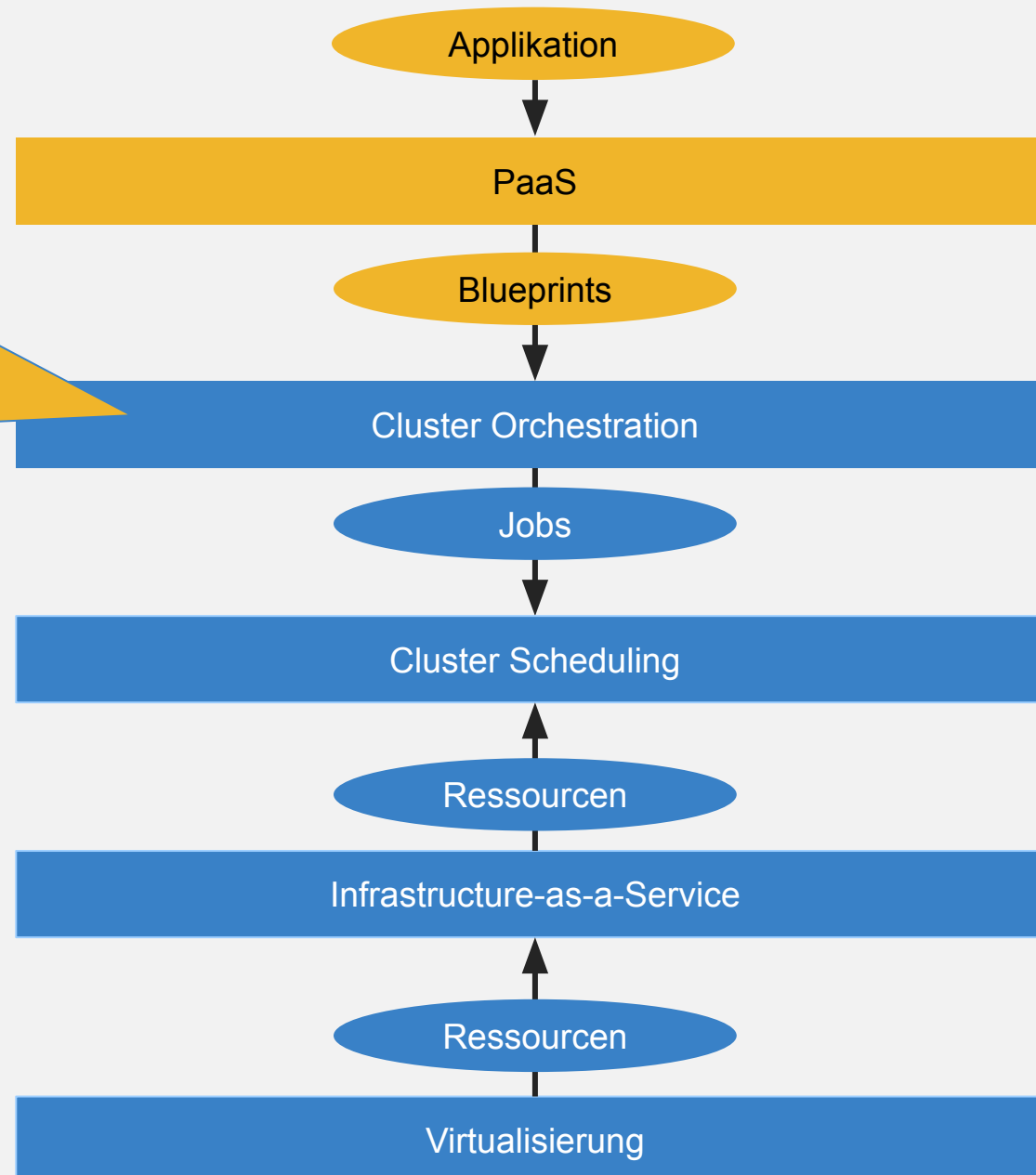


QA|WARE

NIST: The capability provided to the consumer is to **deploy onto the cloud infrastructure** consumer-created or acquired applications created using programming **languages, libraries, services, and tools supported by the provider**. The consumer **does not manage** or control the underlying cloud infrastructure including network, servers, operating systems, or storage, but **has control over the deployed applications** and possibly configuration settings for the application-hosting environment.

Forrester: A complete application platform for multitenant cloud environments that includes development tools, runtime, and administration and management tools and services. PaaS **combines an application platform with managed cloud infrastructure services**.

Big Picture



Hier ist man bereits bei 80% der technischen Basis einer PaaS. Was noch fehlt:

- Wiederverwendung von Infrastruktur / APIs
- Komfort-Dienste für Entwickler

IaaS vs. PaaS

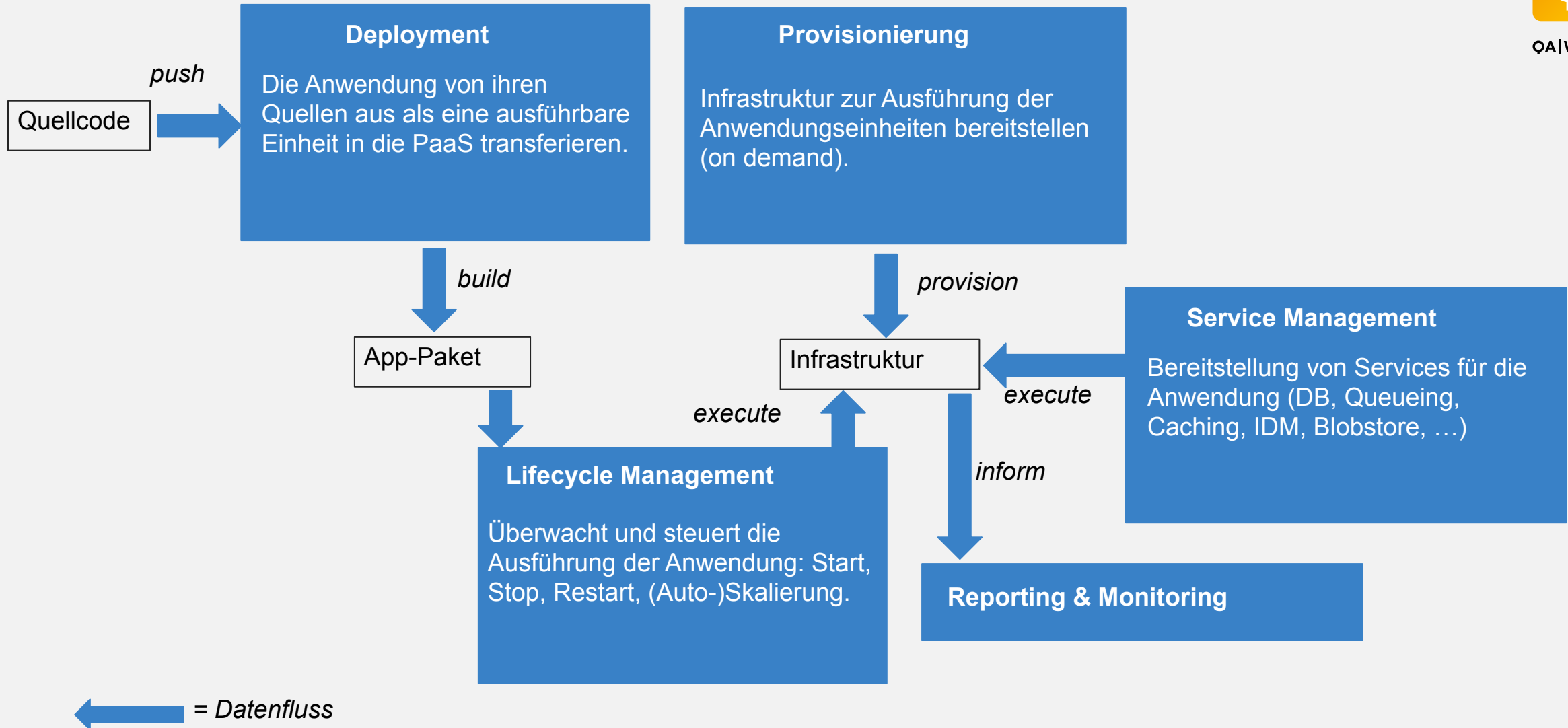


QA|WARE

On-site	IaaS	PaaS	SaaS
Applications	Applications	Applications	Applications
Data	Data	Data	Data
Runtime	Runtime	Runtime	Runtime
Middleware	Middleware	Middleware	Middleware
O/S	O/S	O/S	O/S
Virtualization	Virtualization	Virtualization	Virtualization
Servers	Servers	Servers	Servers
Storage	Storage	Storage	Storage
Networking	Networking	Networking	Networking

-  You manage
-  Service provider manages

Die funktionalen Bausteine einer PaaS Cloud



Beispiel: Cloud Foundry

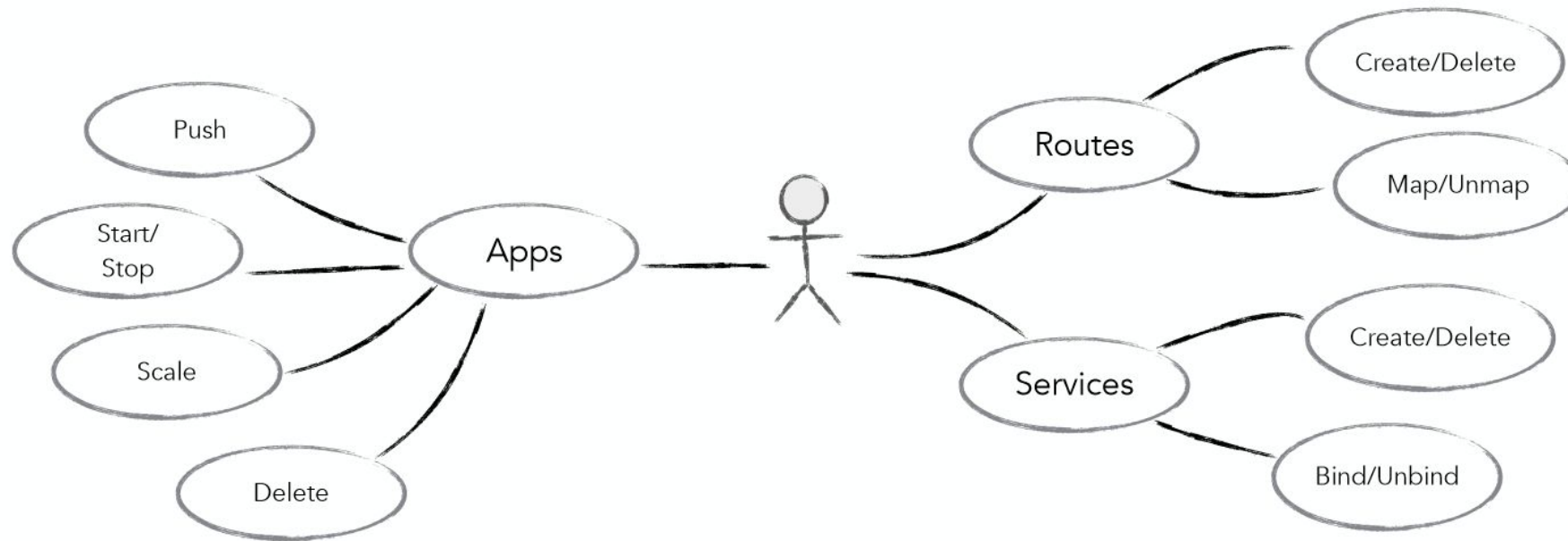


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Als Denkanstoss



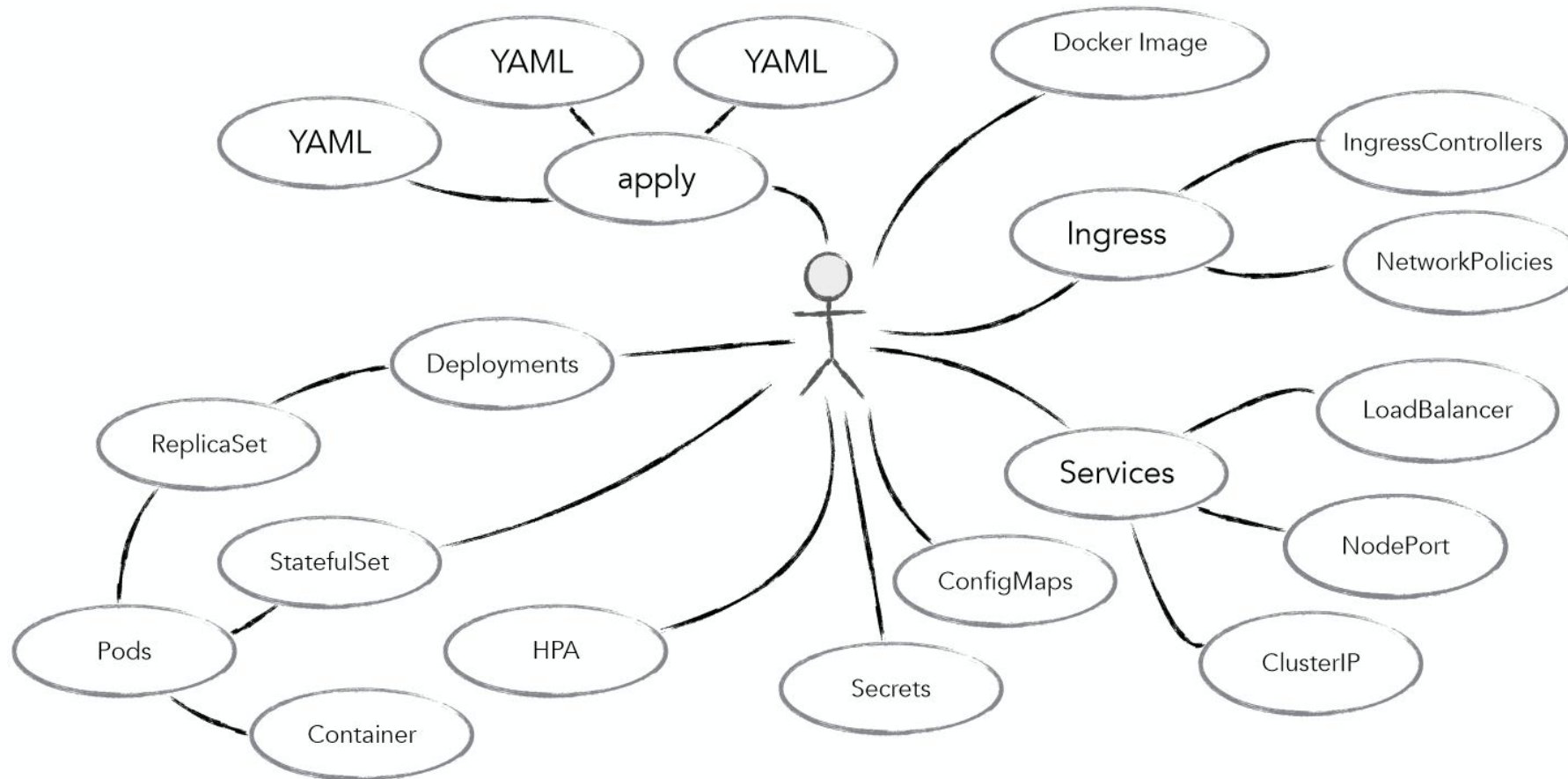
Minimal Concepts



Vergleich zu Kubernetes



Minimal Concepts

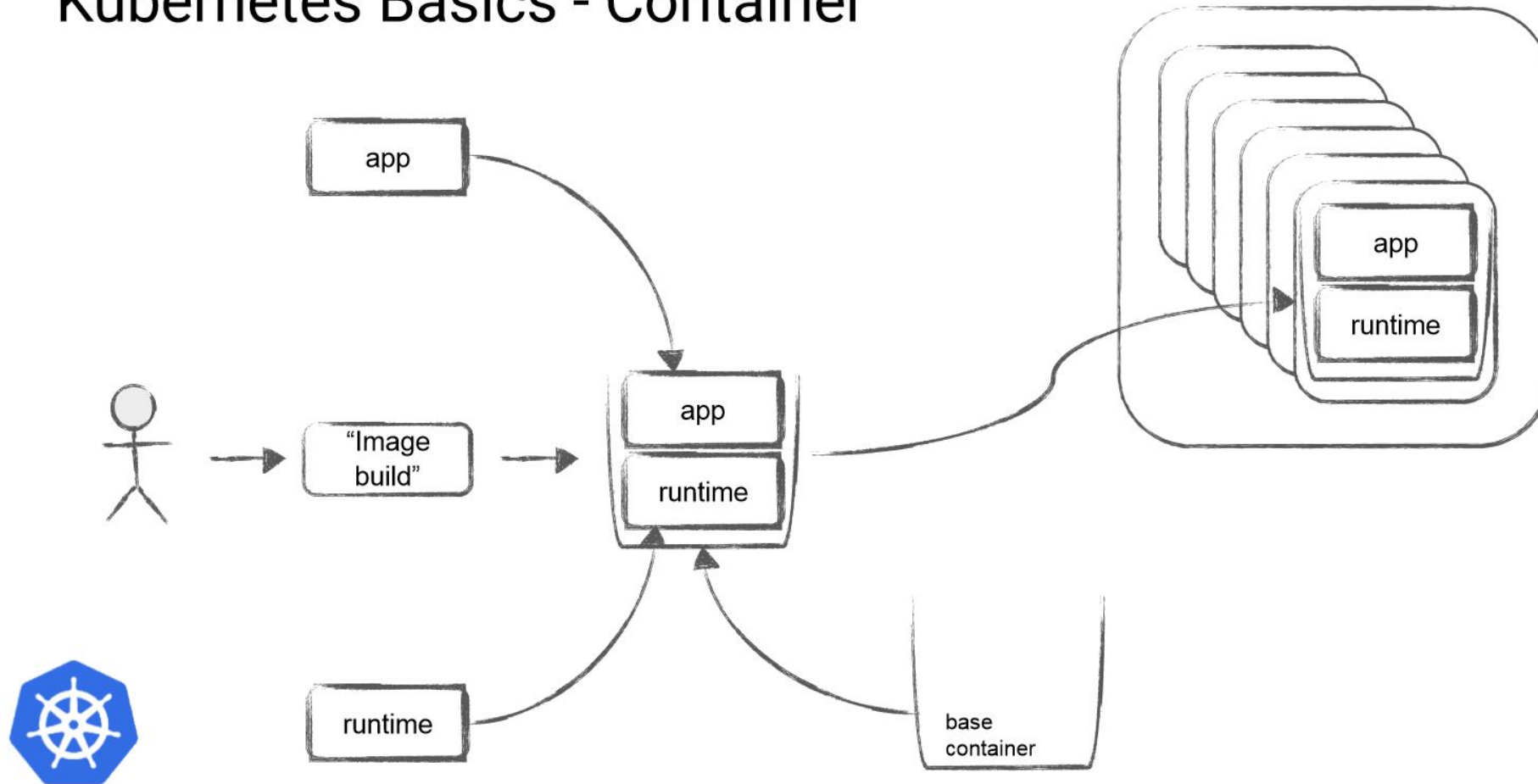


Kubernetes Workflow



QAWARE

Kubernetes Basics - Container

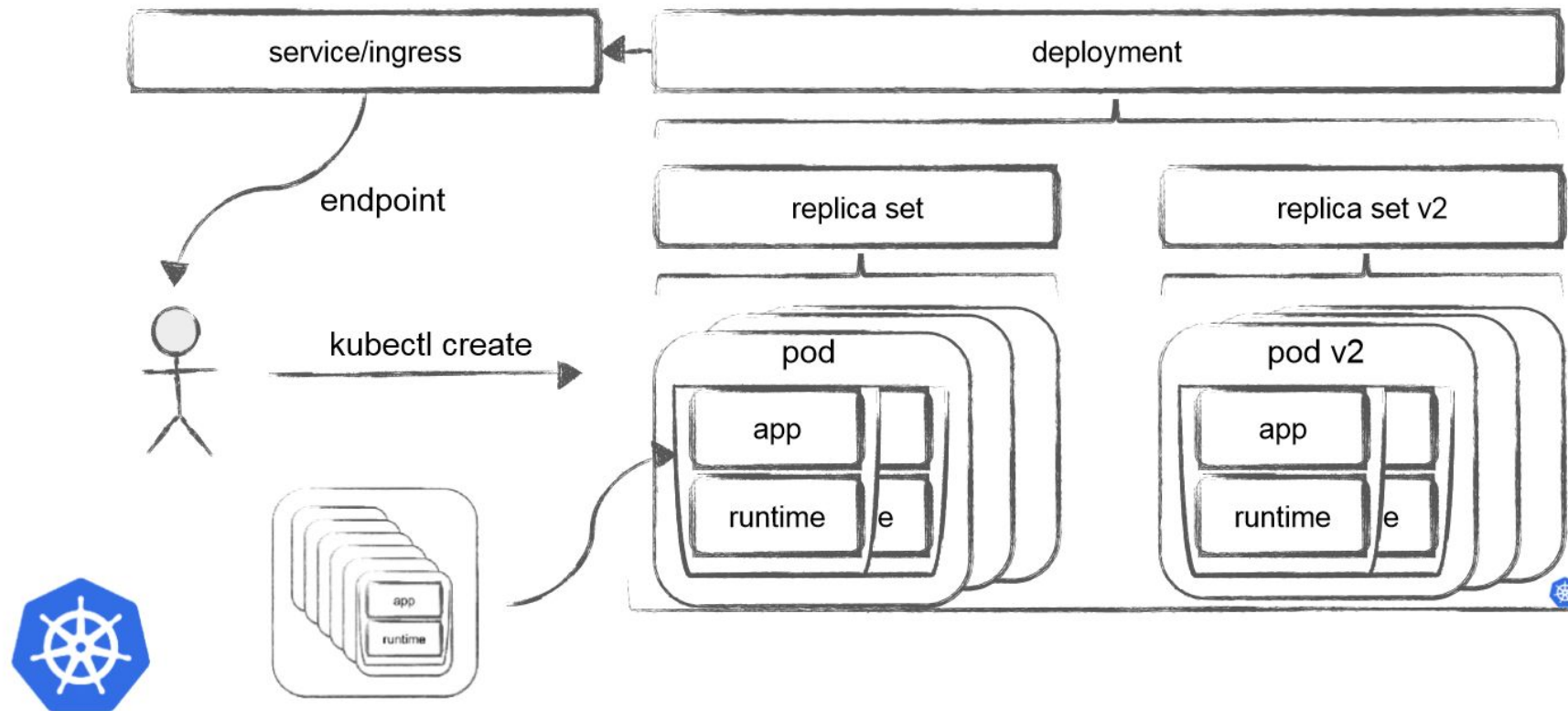


Kubernetes Workflow



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Kubernetes Basics - Orchestration



- Voller Heroku/Vercel-Komfort (Git Push —> Live)
- Läuft überall (eigener VPS, Bare Metal, On-Premise, Cloud)
- Open Source, kostenlos und managt Apps, DBs & SSL vollautomatisch via Docker
- Auch als managed Variante

<https://github.com/coollabsio/coolify>



Custom properties

☆ 57.4k stars

👁 180 watching

🔗 4.8k forks



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Private PaaS

Private PaaS - Übersicht

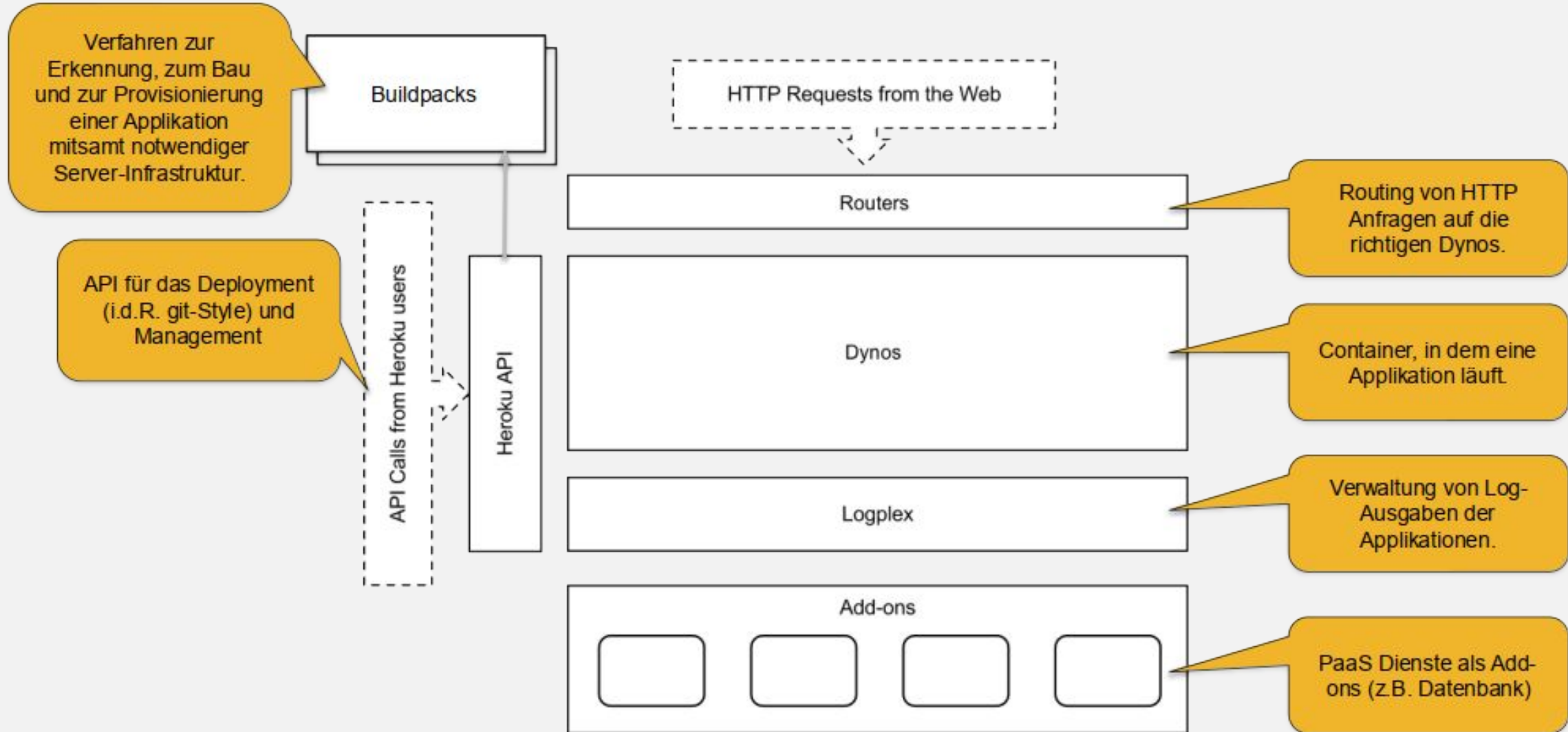
- ~~Flynn~~ (<https://github.com/flynn/flynn>, basiert auf Docker, abandoned)
- ~~DEIS~~ Hefhy (<https://github.com/teamhephy/workflow>, vergleichbar, originales Team wurde von Microsoft acquired, abandoned)
- OpenShift (<https://www.openshift.com>, PaaS mit Schwerpunkt JEE von Red Hat)
- CloudFoundry (<http://www.cloudfoundry.org>, produktionserprobte PaaS von Pivotal mit breiter Unterstützung aus der Industrie)
- ~~Stackato~~ HP Helion (Private PaaS von ActiveState, kommerziell, eingestellt).
- PaaSSTA (<https://github.com/yelp/paasta>). Open-Source, auf Basis von Mesos oder Kubernetes und Marathon, von Yelp
- Coolify (<https://github.com/coollabsio/coolify>)
- ~~Mantl~~ (<http://mantl.io>). Open-Source private PaaS auf Basis von Mesos von Cisco (abandoned)
- Dokku (<https://github.com/dokku/dokku>, Docker-powered Mini-Heroku (in Bash))
- ...

Heroku

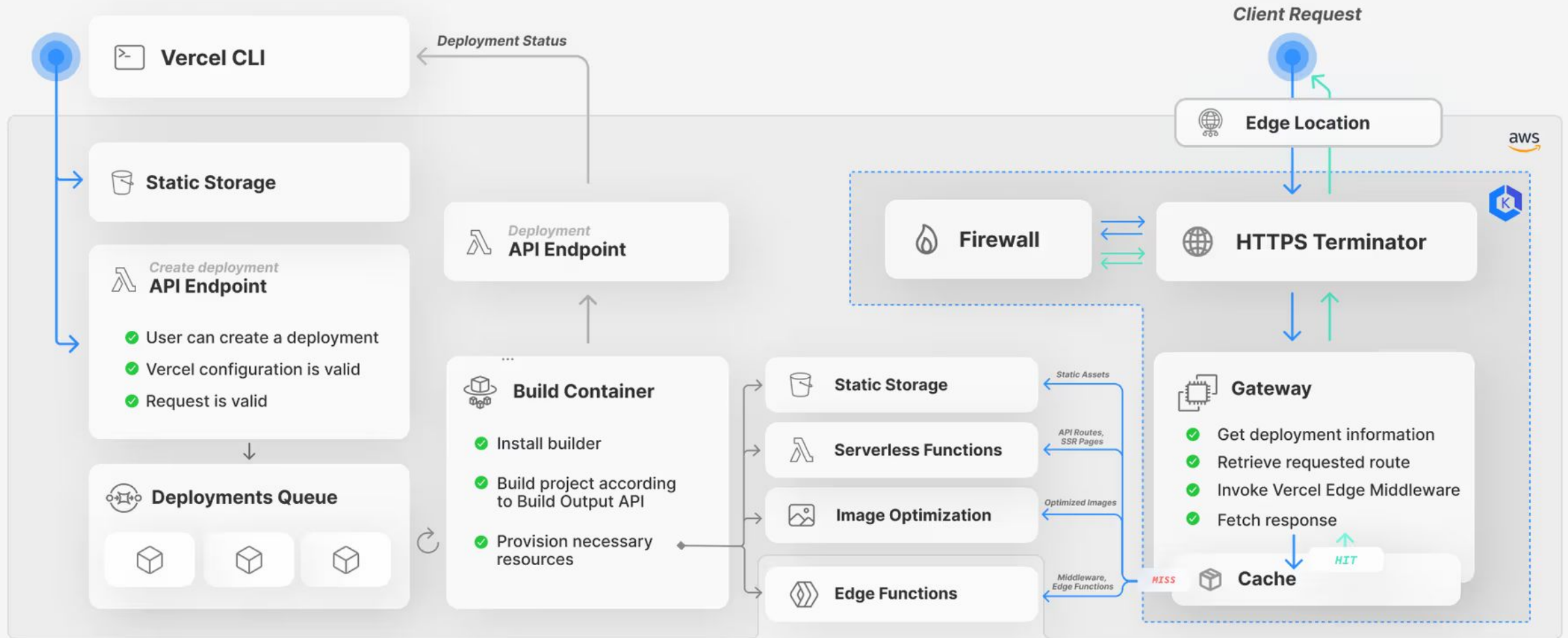
- Eine der ersten PaaS-Plattformen
- Erstes Auftreten 2007
- 2010 von Salesforce gekauft
- Initialer Fokus auf Ruby, mittlerweile wird auch Java, Node.js, Go, Scala, Clojure, Python und PHP unterstützt
- Früher kostenloses Starterangebot inklusive PostgreSQL, jetzt 5\$/Monat
- GitOps-Ansatz: Deployments werden über Git gesteuert



Architektur von PaaS - Beispiel Heroku



Architektur von PaaS - Beispiel Vercel



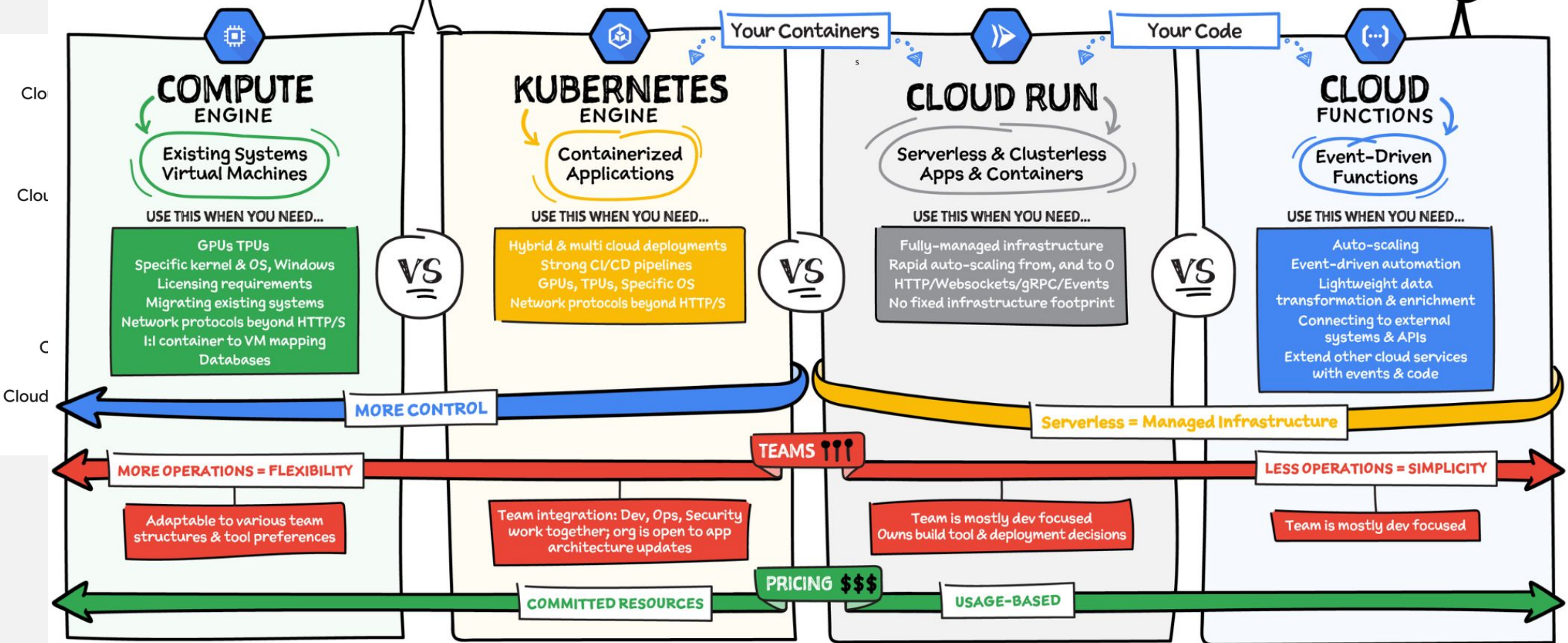


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PaaS in der Public Cloud

Public PaaS: GCP Cloud Run

@PVERGADIA



Google App Engine (GAE)

- Vollgemanagtes PaaS: Kein Server-Setup, automatische Skalierung (bis auf 0)
- Pay-as-you-go: Kostenlose Basis-Quote, Abrechnung strikt nach Verbrauch
- Polyglott: Nativer Support für Java 17/21+, Node.js, Go, Python, .NET
- Modernes Tooling: Git-Push & Deployment via GitHub Actions / IntelliJ (VS Code)



Google App Engine im Überblick



What is App Engine?

FULLY MANAGED SERVERLESS COMPUTE

to build & deploy low latency, highly scalable applications



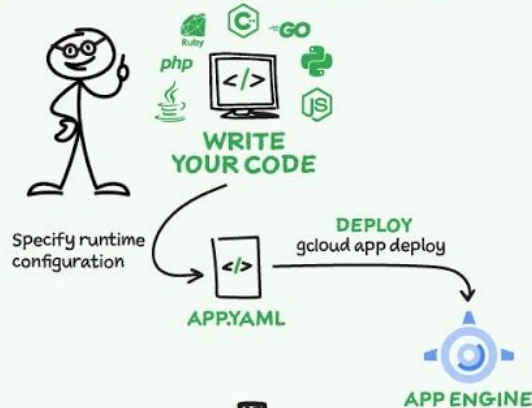
USE APP ENGINE WHEN....

- Web traffic requires minimal latency
- Web framework that supports routes
 - HTTP methods
 - Custom domains

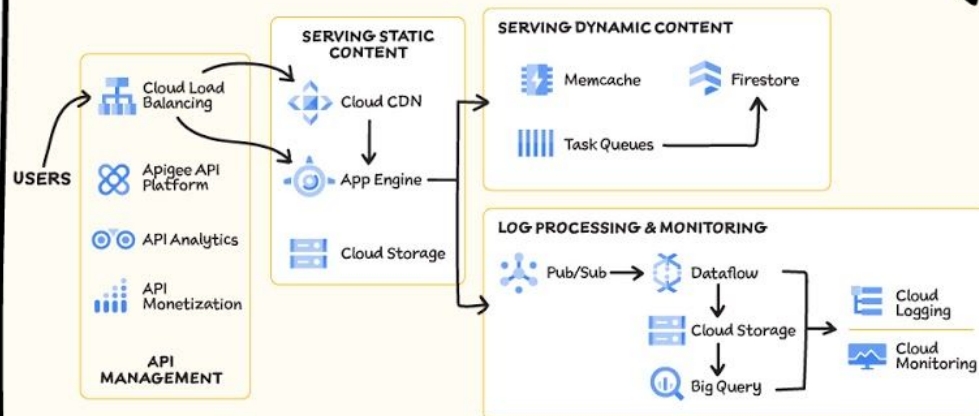
FEATURES

- Autoscaling from 0-many
- Application versioning
- Multiple language support
- Supports traffic splitting across versions for A/B testing
- Container support for custom runtimes
- Custom Domain & managed TLS certificate
- Static file serving

How does it Work? ?



Modern web applications & mobile back ends



*Choosing App Engine Environment



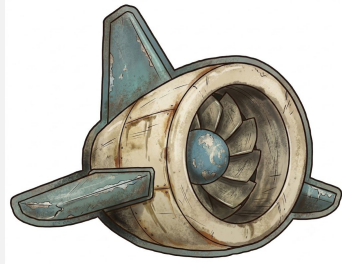
STANDARD

FLEX

Runtime environment for supported language	RUNTIME	Custom runtimes any library/framework
Application instances run in a sandbox		Application instances run within Docker containers on Compute Engine
App with sudden & extreme traffic spikes	TRAFFIC PATTERNS	Apps with consistent traffic, regular fluctuations
Scale from 0-many	SCALE TO ZERO	Scale from 1 to many
Seconds	DEPLOYMENT TIME	Minutes
Depends on runtime	MAX REQUEST TIMEOUT	60 mins
No	WEBSOCKET SUPPORT	Yes

Quelle

Google App Engine Gen1 vs Gen2



Evolution zu offenen Standards



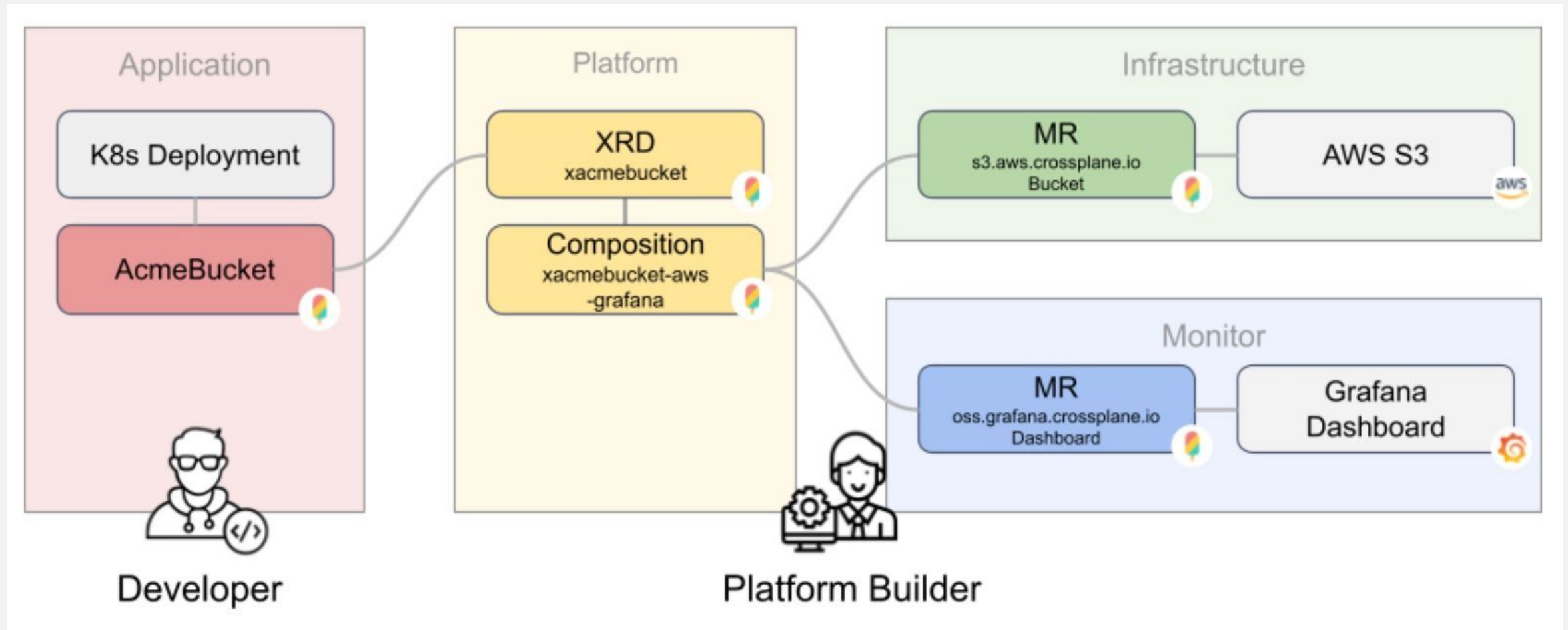
Generation 1	Generation 2
Sandbox Fessel: (kein Schreibzugriff / keine eigenen Threads)	Die offene Laufzeit: Volle Freiheit (normale Container-Umgebung im Hintergrund, /tmp-Schreibzugriff, Threads erlaubt).
Proprietäre APIs: Zwangsnutzung von Google-eigenen Diensten (GAE Datastore, GAE Memcache, Blobstore)	Cloud-Native Dienste: Nutzung standardisierter Google Cloud Produkte (Cloud Firestore, Cloud Storage, Memorystore/Redis).
Code Lock-in: Code musste mit speziellen Google-Bibliotheken geschrieben werden und lief nur bei Google.	Portabel (Idiomatisch): Stinknormaler Standard-Code (z. B. eine reine Spring Boot App). Läuft auf GAE, lokal oder bei AWS.
Alte Runtimes: Festgefroren auf alten Sprachversionen (z. B. Java 8, Python 2.7).	Moderne Runtimes: Java 17/21+, Python 3+, Node.js, Go.



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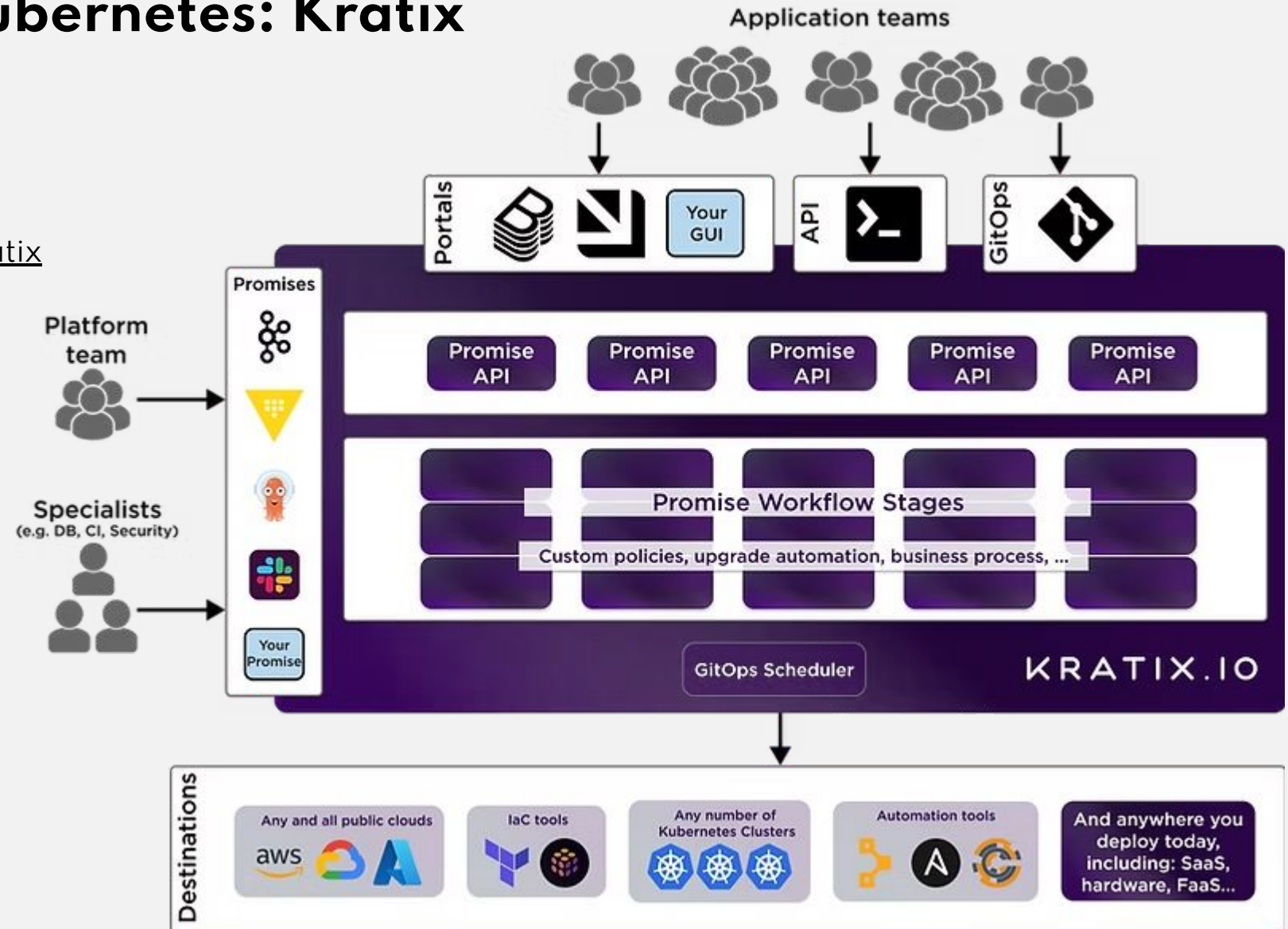
PaaS Frameworks/Bausteine

PaaS-ification von Kubernetes: Crossplane



PaaS-ification von Kubernetes: Kratix

- open-source:
<https://github.com/syntasso/kratix>
- Framework für IDP
(Internal Developer Platform)



No-Code PaaS über AI

I'm Amazon Q, your AI assistant. Ask me anything to start a conversation. I'll give you answers using information from your company. Let's get started!

Draft a press release and a blog

Summarize key trends from our sales pipeline

Create an app to generate our weekly newsletter

Create a password generator in python

To create a password generator, I would:

- Import the random and string modules to help generate random characters
- Define a generate_password function that takes in a length parameter and chooses characters from an approved set
- Call the function with a length of 10 and print the result

Would you like me to write the necessary code?

`</>` Write code



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Exkurs: Internal Development Platforms

CNCF Landscape



QAWARE

Cloud Native Landscape

EXPLORE GUIDE STATS

Filters

GROUP

Projects and productsMembersCertified partners and providersServerlessWasmAI Native

VIEW MODE

GridCard

ZOOM

-

+

Type (/) to search items

🔍

📄

📌

🔗

Application Definition & Image Build

Database

Continuous Integration & Delivery

Streaming & Messaging

Scheduling & Orchestration

Service Proxy

Remote Procedure Call

API Gateway

Service Mesh

Coordination & Service Discovery

Cloud Native Storage

Container Runtime

Cloud Native Network

Security & Compliance

Automation & Configuration

Container Registry

Key Management

dapr

CNCF GRADUATED

HELM

CNCF GRADUATED

Artemis HUB

CNCF INCUBATING

Backstage

CNCF INCUBATING

Buildpacks.io

CNCF INCUBATING

KubeVela

CNCF INCUBATING

KubeVirt

CNCF INCUBATING

MICROCKS

CNCF INCUBATING

OPERATOR FRAMEWORK

CNCF INCUBATING

Dayson

CNCF INCUBATING

CAPIVUL

CNCF INCUBATING

Devfile.io

CNCF INCUBATING

DevSpace

CNCF INCUBATING

Fabric8

CNCF INCUBATING

Gelfy

CNCF INCUBATING

Gitpod

CNCF INCUBATING

Kopio

CNCF INCUBATING

Kaniko

CNCF INCUBATING

Kubernetes

CNCF INCUBATING

K8s

CNCF INCUBATING

K8s

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K8s

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Kui

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Lagoon

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KEDA

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Knative

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kubernetes

CNCF GRADUATED

FLUID

CNCF INCUBATING

KARMADA

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Kubeflow

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VOLCANO

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WasmCloud

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Lima

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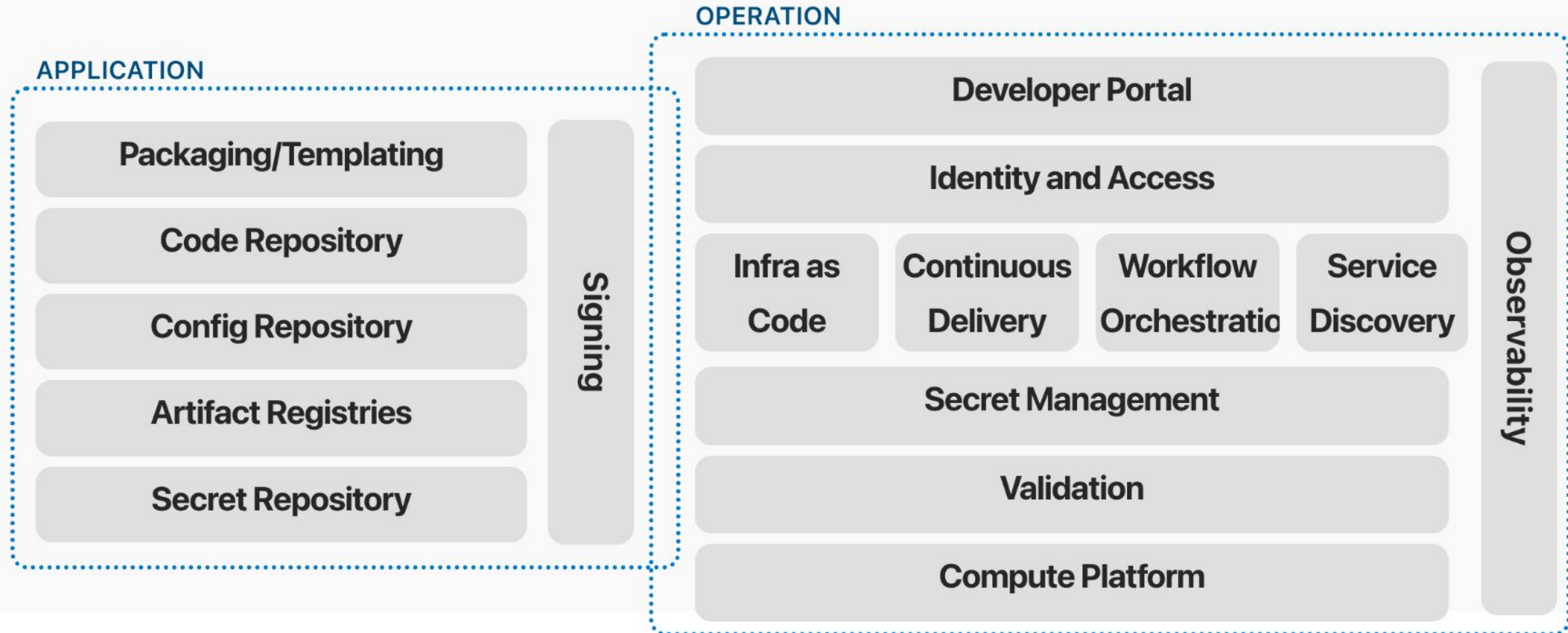
Cilium

CNCF GRADUATED

CNOE (pronounced Kuh.noo): Cloud Native Operational Excellence - IDP Referenzimplementierung



QA|WARE

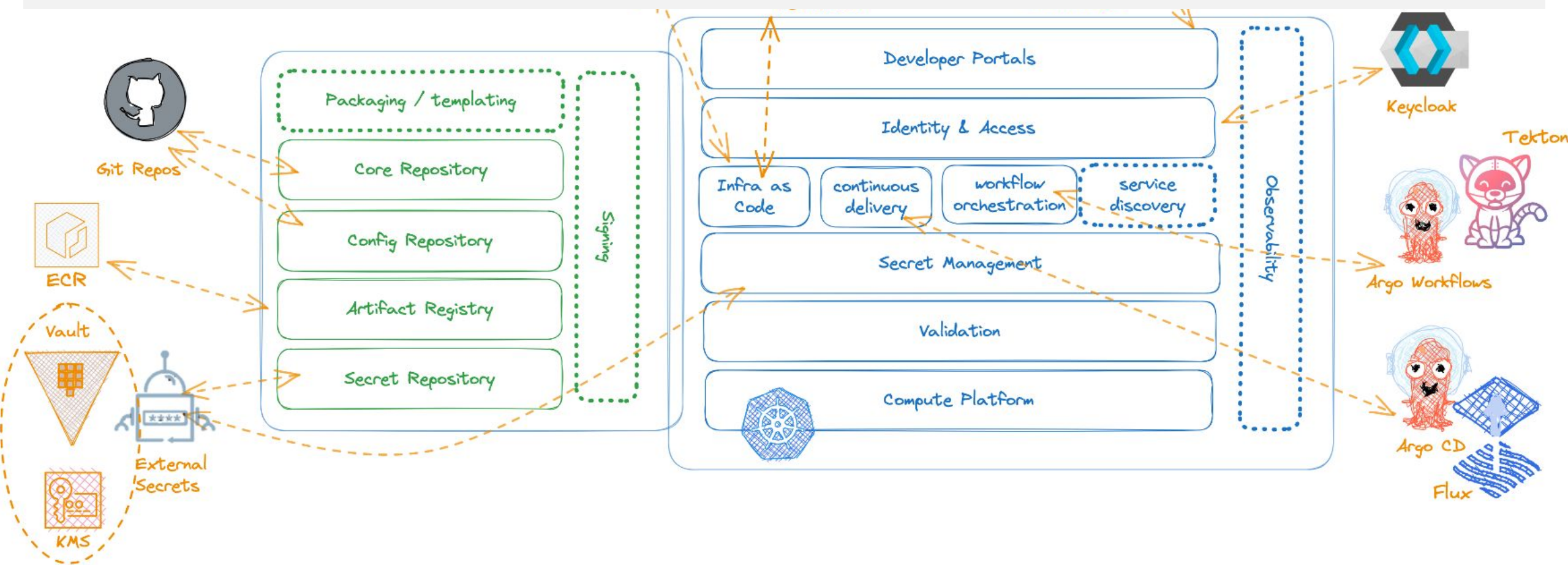


CNOE: Cloud Native Operational Excellence

IDP Referenzimplementierung



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Backstage als IDP-Enabler



QA|WARE

 Backstage

Search

Home

Catalog

APIs

Docs

Explore

Tech Radar

Cost Insights

GraphQL

Settings

Backstage Catalog

Kind
Component

Type
all

PERSONAL
Owned 0
Starred 0

BACKSTAGE
All 15

OWNER

All components (15)

Filter

NAME	SYSTEM	OWNER	TYPE	LIFECYCLE	DESCRIPTION	TAGS	ACTIONS
artist-lookup	artist-engagement-portal	team-a	service	experimental	Artist Lookup	java, data	
backstage		cncf	library	experimental	Backstage is an...		
backstage-demo		backstage/mainainers	website	experimental	An example...		
petstore		Team C	service	experimental	[The...		
playback-order	audio-playback	Guest User	service	production	Playback Order	java, playback	
playback-sdk	audio-playback	Team C	library	experimental	Audio and video...		
podcast-api	podcast	Team B	service	experimental	Podcast API	java	
queue-proxy	podcast	Team B	website	production	Queue Proxy	go, website	
searcher		Guest User	service	production	Searcher	go	

Backstage als IDP-Enabler



QA|WARE

Create a New Component | Temp | x

localhost:3000/create?filters%5Bkind%5D=template&filters%5Buser%5D=all

Backstage

Search

Home

APIs

Docs

Create...

Tech Radar

Create a New Component

Create new software components using standard templates

Available Templates

Search

PERSONAL

★ Starred 0

MY COMPANY

All 3

CATEGORIES

TAGS

Templates

documentation

Documentation Template

DESCRIPTION

Create a new standalone documentation project

OWNER

backstage/techdocs-core

TAGS

recommended techdocs mkdocs

CHOOSE

website

React SSR Template

DESCRIPTION

Create a website powered with Next.js

OWNER

web@example.com

TAGS

recommended react

CHOOSE

service

Spring Boot gRPC Service

DESCRIPTION

Create a simple microservice using gRPC and Spring Boot Java

OWNER

service@example.com

TAGS

recommended java grpc

CHOOSE

IDP-Komponenten: Humanitec: “Score.yaml”



```
score.yaml
apiVersion: score.dev/v1b1

metadata:
  name: python-service

containers:
  python-service:
    image: python
    variables:
      CONNECTION_STRING: postgresql://${resources.db.user}:${resources.db.password}@${resources.db.host}:${resources.db.port}/${resources.db.name}

resources:
  db:
    type: postgres
  storage:
    type: s3
  dns:
    type: dns
```

